The Extremal Principle — problem 3.34

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Let E be a finite set of points in 3-space with the following properties :
(a) E is not coplanar. (b) No three points of E are collinear.
Prove : Either there are five points in E, which are vertices of a convex pyramid the interior of which is free of points of E, or there exists a plane, which contains exactly three points of E.